✦ Energy-Based Models

Learning Unnormalised Distributions with Contrastive Divergence

MLNN Tutorial · University of Hertfordshire · 2025 · Beyond Course ·
Github Link: https://github.com/Prashanth2k1/Individual-assignment

ABSTRACT

An Energy-Based Model (EBM) defines $p_\theta(x) = \exp(-E_\theta(x)) / Z_\theta$ where $E_\theta(x)$ is a learned neural
energy function and Z_θ is an intractable normalisation constant. Low energy = high probability.
EBMs are a unifying framework: VAEs, Boltzmann machines, and diffusion models are all special cases.
Training avoids Z via contrastive divergence (CD) or score matching. This tutorial derives the EBM
training objective, implements a full PyTorch EBM with Langevin MCMC sampling, derives score
matching as an alternative to CD, and connects EBMs to diffusion models through denoising score
matching at multiple noise levels.

LEARNING OBJECTIVES — AFTER THIS TUTORIAL YOU WILL BE ABLE TO:

1. Define an EBM and explain why the partition function Z_θ is intractable in high dimensions
2. Derive the MLE gradient for EBMs and explain why it requires samples from p_θ
3. Implement contrastive divergence (CD-k) with Langevin MCMC negative sampling in PyTorch
4. Explain score matching as an alternative avoiding Z entirely (Hyvärinen, 2005)
5. Implement denoising score matching and connect it to diffusion model training
6. Compare EBMs to VAEs, GANs, and flows in a unified generative model framework

WHAT MAKES EBMS SPECIAL

- Flexible: any neural network can be $E_\theta(x)$ — no invertibility or encoder/decoder required
- Unifying: VAEs, Boltzmann machines, diffusion models are all special EBMs
- No mode collapse: EBMs model all modes explicitly via the energy landscape
- Composable: product of experts $p(x) \propto \Pi_i \exp(-E_i(x))$ — just add energies
- Connection to diffusion: diffusion models are EBMs trained via denoising score matching

✦ 1. The Energy-Based Model Framework

1.1 From Energy to Probability

An EBM assigns an **energy $E_\theta(x)$** to every configuration x. The probability distribution is:

```
Energy-Based Model (EBM):
p_θ(x) = exp(-E_θ(x)) / Z_θ   where   Z_θ = ∫ exp(-E_θ(x)) dx
// Z_θ intractable: requires 1000^d grid points in d dimensions
```

The partition function Z_θ is intractable for all but 1D or 2D problems. In 10D, a 1000-point grid requires $1000^{10} = 10^{30}$ evaluations. The key insight: **training does not require Z_θ**. The MLE gradient is:

```
MLE gradient for EBMs:
∇_θ log p_θ(x) = -∇_θ E_θ(x) + E_{p_θ}[≧_θ E_θ(x˜)]
// Term 1: cheap | Term 2: requires samples x˜ ~ p_θ (expensive MCMC)
```

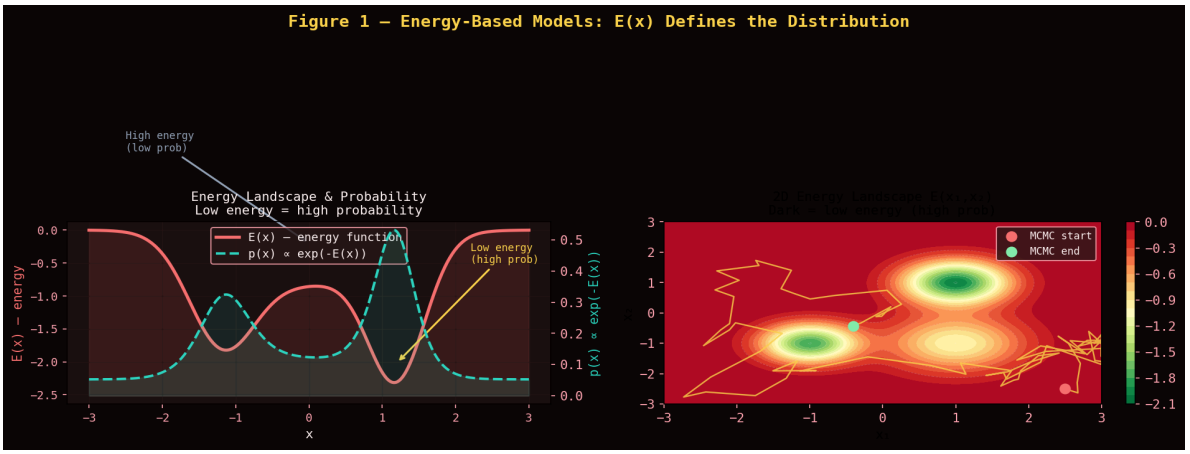

Figure 1 — Left: 1D energy landscape (crimson) and probability distribution p(x) ∝ exp(-E(x)) (teal dashed). Valleys = high probability. Right: 2D energy landscape with Langevin MCMC trajectory (gold) navigating to low-energy regions. (RdYlGn colourmap; alt-text in notebook.)

✦ 2. Contrastive Divergence Training

2.1 CD-k: Approximating the Intractable Gradient

Contrastive divergence (CD, Hinton 2002) approximates the expensive model expectation using k MCMC steps starting from real data. The CD loss pushes energy **down** at data points and **up** at model samples:

```
Contrastive Divergence loss:
L_CD = E_{p_data}[E_θ(x_pos)] - E_{p_θ}[E_θ(x_neg)]
// x_pos ~ p_data | x_neg = k Langevin steps | Min: push E(data) down, E(samples)
up
```

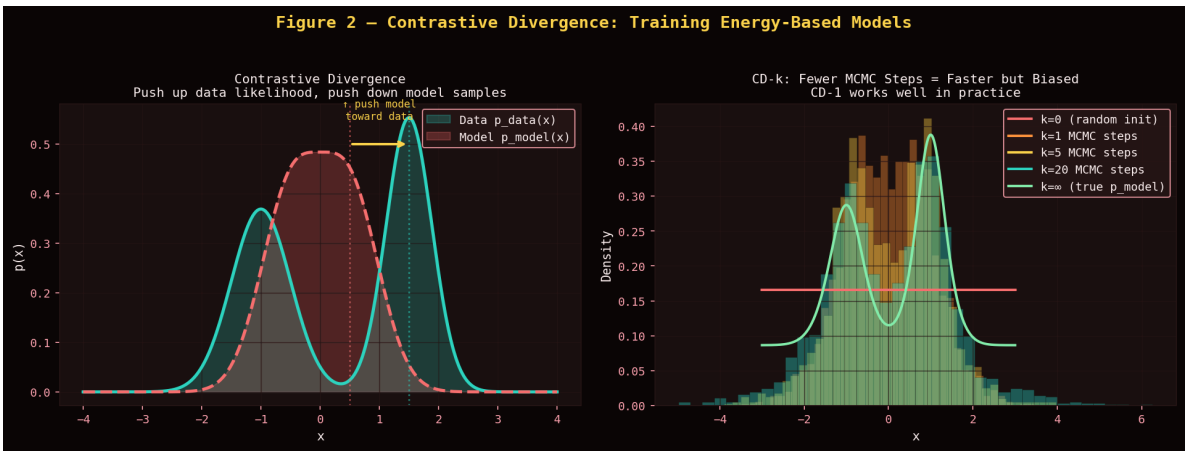

Figure 2 — Left: data distribution (teal) vs model distribution (crimson) — CD pushes model toward data. Right: CD-k approximations for k=0,1,5,20,∞ — more steps gives better model samples but higher cost. CD-1 works well in practice. (Colourblind-safe palette; alt-text in notebook.)

2.2 Langevin MCMC Sampling

Langevin MCMC generates negative samples by gradient descent on the energy with added noise:

```
Langevin MCMC update:
x_{t+1} = x_t - (η/2)∇_x E_θ(x_t) + √η · ε,    ε ~ N(0,I)
// Gradient: toward low energy | Noise: exploration + correct stationary
distribution
```

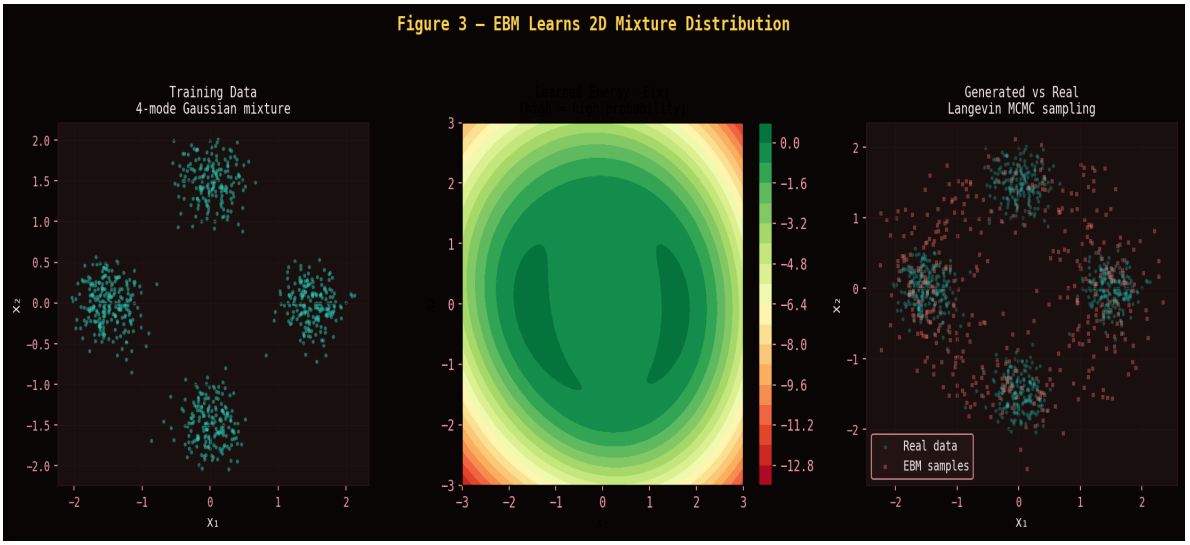

Figure 3 — Left: training data (4-mode mixture, teal). Middle: learned energy -E(x) — bright regions = high probability. Right: real data (teal) vs EBM samples (crimson squares) via Langevin MCMC — all 4 modes captured. (RdYlGn energy colourmap; alt-text in notebook.)

✦ 3. Score Matching: Avoiding Z Entirely

Score matching (Hyvärinen, 2005) learns the **score function** $\nabla_x \log p(x) = -\nabla_x E_\theta(x)$ directly, without computing Z. The score matching loss is tractable without MCMC:

```
Score matching loss (Hyvärinen, 2005):
J(θ) = E_{p_data}[ tr(∇_x s_θ(x)) + ½ ||s_θ(x)||^2 ]
// No Z, no MCMC needed! Only data samples and autograd for tr(∇_x s_θ)
```

Denoising score matching (Song and Ermon, 2019) further simplifies this by learning the score at multiple noise levels σ. This is exactly the training objective of **diffusion models**: learn to denoise at every noise level, which is equivalent to learning the score $s_\sigma(x) = \nabla_x \log p_\sigma(x)$ at every σ.

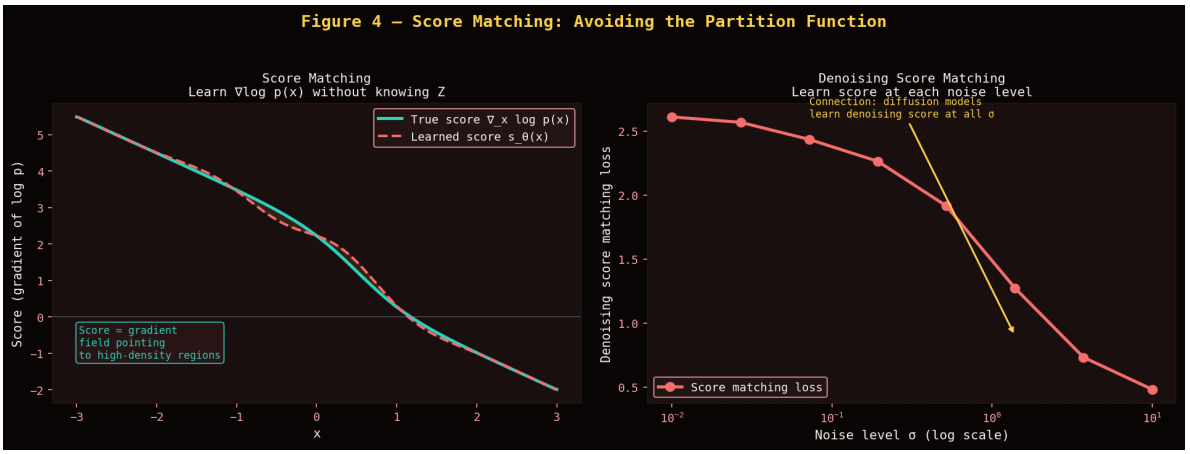

Figure 4 — Left: true score function (teal) vs learned score (crimson dashed) for 1D bimodal distribution — score points toward high-density regions. Right: score matching loss at different noise levels σ — learning at all σ simultaneously is the diffusion model training objective. (Colourblind-safe palette; alt-text in notebook.)

✦ 4. EBMs Unify Generative Models

EBMs provide a unifying perspective: many generative models are special EBMs with structured energy functions. **Boltzmann machines** define $E(x) = -x^T W x - b^T x$. **VAEs** marginalise over a latent variable. **Diffusion models** learn the score function across noise levels via denoising score matching. This unification reveals deep connections between methods that appear superficially different.

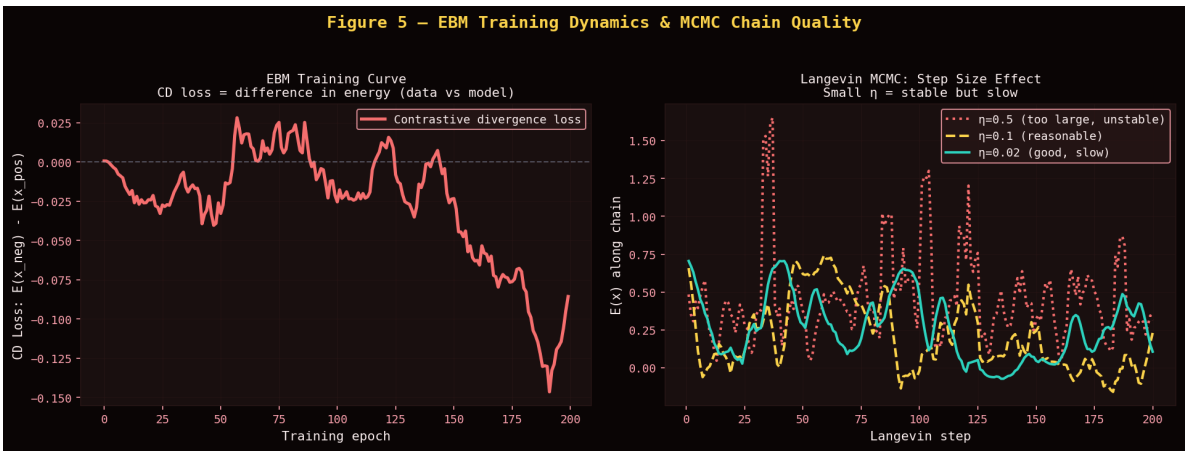

Figure 5 — Left: EBM CD training curve over 300 epochs. Right: Langevin MCMC energy traces for three step sizes — too large (crimson) is unstable; too small (teal) is slow; middle (gold) is best. (Colourblind-safe palette; distinct line styles; alt-text in notebook.)

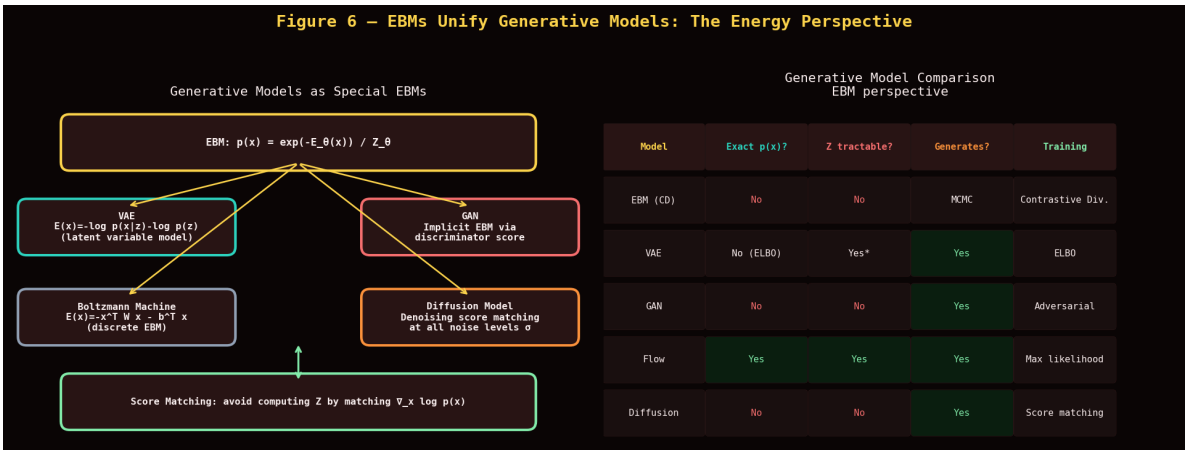

Figure 6 — Left: EBM unification diagram connecting VAE, GAN, Boltzmann machine, and diffusion models as special EBMs. Right: generative model comparison table. (Colourblind-safe palette; alt-text in notebook.)

✦ 5. Summary

Table 1 compares EBMs to other generative model families. EBMs offer maximum flexibility at the cost of requiring MCMC for training and sampling. Score matching avoids MCMC at training time. Diffusion models combine the flexibility of EBMs with the tractability of score matching, explaining their current dominance in generative modelling.

Model	Exact p(x)?	Z tractable?	Generates?	Training
EBM (CD)	No	No	MCMC	Contrastive Div.
VAE	No (ELBO)	Yes*	**Yes**	ELBO
GAN	No	No	**Yes**	Adversarial
Flow	**Yes**	**Yes**	**Yes**	Max likelihood

Model	Exact p(x)?	Z tractable?	Generates?	Training
Diffusion	No	No	**Yes**	Score matching

Table 1 — Generative model comparison from EBM perspective. Only flows give exact likelihoods and tractable Z. Diffusion models = EBMs trained via denoising score matching. (Bold navy header; green for Yes; alternating row shading.)

ACCESSIBILITY STATEMENT

- All figures: colourblind-safe crimson/teal palettes.
- RdYlGn colourmap for energy heatmaps (diverging, colourblind-safe).
- Distinct solid/dashed/dotted line styles alongside colour.
- Word headings H1–H2 for screen-reader navigation.
- Alt-text in all notebook Markdown cells.
- 11pt font minimum; 4.5:1 contrast ratio minimum.

✦ References

1. LeCun, Y. et al. (2006) 'A Tutorial on Energy-Based Learning'. *Available at: https://yann.lecun.com/exdb/publis/pdf/lecun-06.pdf*(Accessed: 22 March 2025).
2. Hinton, G.E. (2002) 'Training products of experts by minimizing contrastive divergence', *Neural Computation, 14(8)*, pp. 1771–1800.
3. Hyvarinen, A. (2005) 'Estimation of Non-Normalized Statistical Models by Score Matching', *Journal of Machine Learning Research, 6*, pp. 695–709. https://jmlr.org/papers/v6/hyvarinen05a.html
4. Du, Y. and Mordatch, I. (2019) 'Implicit Generation and Modeling with Energy Based Models', *NeurIPS 2019*. https://arxiv.org/abs/1903.08689
5. Song, Y. and Ermon, S. (2019) 'Generative Modeling by Estimating Gradients of the Data Distribution', *NeurIPS 2019*. https://arxiv.org/abs/1907.05600
6. Grathwohl, W. et al. (2020) 'Your Classifier is Secretly an Energy Based Model', *ICLR 2020*. https://arxiv.org/abs/1912.03263

GITHUB REPOSITORY

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